AMSSE-Net: Adaptive Multiscale Spatial–Spectral Enhancement Network for Classification of Hyperspectral and LiDAR Data

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***Abstract*— With the abundant emergence of remote sensing (RS) data sources, multimodal remote sensing observation has become an active field. Extracting valuable information from multimodal data has the potential to make a significant contribution to applications such as urban planning and monitoring. However, existing studies are deficient in extracting spectral and spatial features from hyperspectral (HS) remote sensing data. Meanwhile, the method of fusing multimodal features has limitations and poses a challenge to the convergence of the model loss function, which increases the complexity of the network model optimization process. Therefore, this article proposes an adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and light detection and ranging (LiDAR) data called AMSSE-Net. First, we perform deep mining of spectral features in hyperspectral images (HSIs) by the involution operator. The main idea is to take full advantage of the involution operator in characterizing spectral features by using the property that the convolution kernel shares the feature channels within the group. Furthermore, the multibranching approach is used to extract the multiscale information, and then the spectral–spatial features are formed with the strategy of hierarchical fusion. Meanwhile, we employ three-layer convolution for extracting shallow features from LiDAR data, offering supplementary information. Finally, we propose the "adaptive feature fusion module (AFFM)," an innovative and comprehensive mechanism designed for the fusion of features from diverse sources in multisource data fusion. These dynamically assigned weights guide the selection of the optimal model, which is determined by the joint loss across the three methods, ultimately leading to the generation of an accurate prediction map. This approach not only helps to deeply explore the spectral spatial information in the hyperspectral data, but also fuses the hyperspectral information effectively with the elevation information from the LiDAR data. The expression ability of model features is rapidly improved by adaptive weighting, which in turn enhances the performance and generalization ability of the model. Compared with some existing methods, extensive experiments on three popular HSI and LiDAR datasets show that our proposed AMSSE-Net can achieve better classification performance. The codes will be available at https://github.com/haofeng0003/AMSSE-Net, contributing to the RS community.**

***Index Terms*— Adaptive feature fusion, involution, multimodal data classification, multiscale hierarchical fusion (MHF).**

I. Introduction

HYPERSPECTRAL (HS) remote sensing (RS), also known as imaging spectral remote sensing, is an advanced remote sensing technology for acquiring and processing spectral information of ground objects. Hyperspectral remote sensing collects hyperspectral images (HSIs) that combine the sample's image and spectral information, which contains rich spatial, radiometric, and spectral information [1], [2]. The image information can reflect the size, shape, and other external quality characteristics of the sample, and because different components have different spectral absorption, the image at a particular wavelength will have a more significant reflection of certain features, so remote sensing technology plays a vital role in land cover classification [3], scene classification [4], urban planning [5], target detection [6], and mineral exploration [7] and other Earth observation tasks. More accurate and detailed surface information can be obtained through hyperspectral remote sensing technology, providing more reliable and precise data support for various applications.

In the early studies of HSI classification, most of the methods focused on exploring the role of HSI spectral features in classification, such as support vector machines (SVM) [8], multinomial logistic regression [9], and dynamic or random subspaces. Therefore, in [10], combining standard SVM classification with hierarchical methods was proposed to improve the accuracy of SVM classification. In [11], some improved kernels are proposed to consider the spectral similarity between support vectors to classify hyperspectral data cubes in a way that is superior to using only SVM. Li et al. [12] proposed a new soft sparse polynomial logistic regression (S^2MLR) model for the semi-supervised classification of hyperspectral images using hard and soft labels.

However, the classification maps obtained by these pixel-level classifiers are not satisfactory due to the lack

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of consideration of spatial context. To solve this problem, Gu et al. [13] proposed a novel spatio-temporal spectral kernel method (S2MKL) for hyperspectral image classification, where extended morphological features are used as spatial information and raw spectral features are used as input features, data in different feature spaces are mapped into the same feature space, and then these features are fused using a multikernel learning algorithm. Sun and Zhang [14] used extended multi-attribute profiles (EMAPs) to model the spatial information and proposed a composite kernel classification method combining spectral–spatial and abundance information to improve the accuracy of classification. Li et al. [15] proposed joint learning framework for noise reduction while enhancing semantic content and achieved better classification results. In [16], a framework combining integration learning and composite kernel learning was proposed to classify hyperspectral images by creating a rotation matrix, which is used to generate rotated features for spectral and spatial information, which in turn results in a variety of new spectral and spatial features. Gao et al. [17] proposed a novel multiscale two-branch feature fusion and attention network in response to previous methods focusing on feature extraction through the use of fixed convolutional kernels and layer-by-layer representations as a way to better adapt to features at different scales while taking into account both spectral and spatial information.

Although most spectral space-based classification methods have achieved good performance, they rely heavily on manual or shallow-based descriptions and are prone to lead to the Hughes phenomenon [18], [19]. Moreover, urban and rural areas often have many complex objectives. In particular, different types of terrestrial objects have similar spectra, or the same terrestrial species have a very different spectrum. In this way, it is easy to produce phenomena such as different objects with the same spectrum and the same object with different spectrums, thereby reducing the classification accuracy of the classifier [20]. For example, light detection and ranging (LiDAR) data assisted by an optoelectronic detection system contains precise elevation information, thus providing complementary information to hyperspectral data and greatly enhancing the performance of the network in feature extraction and classification [21]. Zhang et al. [22] proposed HS and LiDAR spatial transformers, aiming to mine the pixel feature embedding relationship globally to capture the spatial and elevation information of HS and LiDAR, respectively, and explore the shared spatial features between the two modalities. To cope with this problem, several studies have been devoted to fusing data from various sensors.

Among various deep learning (DL) models, convolutional neural networks (CNNs) have gained the most attention and have been applied in multiple tasks [23], [24], [25]. For example, Zhang et al. [26] proposed a multiscale patch-to-patch (PToP) CNN for feature extraction, which ensures perfect information and robustness of extracted features under a small training sample size due to its neural weight learning process, which is entirely unsupervised. Xu et al. [27] proposed a dual-tunnel CNN framework to extract spectral–spatial features from HSI and features from LiDAR images, respectively. Zhang et al. [28] also proposed an interleaved perceptual convolutional neural network (IP-CNN) for information fusion based on the dual-tunnel CNN. Both of the above methods used dual-tunnel CNN and achieved good classification performance. However, these two approaches have resulted in an excess of parameters, highlighting the challenge of effectively harnessing this abundance of data. Therefore, Hang et al. [29] proposed to use two coupled CNNs to fuse hyperspectral and LiDAR data to reduce the number of model parameters by training with shared parameters. One CNN is used to learn spectral–spatial features from hyperspectral data, and the other is used to capture elevation information from LiDAR data. And finally, the two convolutional layers are coupled by a parameter-sharing strategy.

The communication and integration of features extracted from diverse modal data are constrained, thereby manifesting notable limitations in terms of semantic relevance [30]. Moreover, the conventional convolution operation performs a weighted average operation on each pixel in the sensory field, which blurs the image and may reduce the discriminability of the center pixel due to the effect of interfering pixels. To address this problem, some improved convolution operations have been introduced, such as dilated convolution and attention mechanism [31]. Hong et al. [32] propose a simple and effective multimodal DL baseline. The deep encoder–decoder network architecture (EndNet) forces the fusion of features while reconstructing multimodal input data to fuse multimodal information better to activate neurons across modes. Fang et al. [33] proposed a spatial–spectral enhancement module (S2EM), which can be regarded as a cross-modal attentional mechanism to enhance the spatial representation of the hyperspectral data and the spectral representation of the LiDAR data through LiDAR features and hyperspectral features, respectively, to facilitate the interaction and understanding multimodal data. In [34], a feature fusion and extraction framework (FusAtNet) is proposed, where feature learning based on "cross-attention" utilizes the attention of one modality (LiDAR) to highlight the features of another modality (HSI). Wu et al. [35] proposed a cross-channel reconstruction module (CCR-Net) with a CNN as the backbone to allow for the efficient exchange of information between modalities. Dong et al. [36] proposed an effective self-guided and cross-guided attention multibranch feature fusion network (MB2FscgaNet) for the joint classification of HSI and LiDAR, using spectral supplementary module (SeSuM) to supplement the discriminative spectral features further and enhance the discriminatory ability of the center pixel.

The recently proposed models treat multiple modalities equally during feature extraction processing. These models aim to efficiently integrate and extract features from different modalities without biasing or favoring any particular modality. For example, Wang et al. [37] designed the multiscale interactive information extraction (MIIE) module to extract multiscale information from two modalities. However, given that HSI data contains richer information than LiDAR data and LiDAR data mainly provides supplementary information to hyperspectral data, more attention needs to be paid to the complementary ability of multisource data in collaborative

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classification tasks and to enhance the semantic relevance of multisource data [38]. Therefore, Wang et al. [39] emphasized the importance of the spectral channel information of HSI and independently measured the contribution of each pixel's spectral channel to the construction of the spectral feature tensor by involution sliding over the spectral channels of the input HSI. Hu et al. [40] proposed a simple but effective parallel transformer model. One transformer acts as a hyperspectral image feature extractor. And the other transformer is responsible for capturing cross-modal interactions. It focuses more on the hyperspectral data, while the Lidar data complements the elevation information for the hyperspectral data.

Although these models can effectively integrate and extract features from different modalities, the redundant information of multisource data also affects the effectiveness of features from heterogeneous data, which reduces the accuracy of joint classification [41]. Therefore, Guo et al. [42] proposed a dual graph convolutional collaborative dense network for HSI and LiDAR classification. Using the Euclidean and cosine maps of HSI, the spectral features with spectral inline are obtained by a dual graph convolution network (GCN), which contains the spectral absolute and relative differences, and the dense network is used to obtain spatial features from LiDAR data. Zhang et al. [43] proposed a structure-optimized transmission framework, structure-optimized transmission network (SOT-Net), to integrate the spectral and geometric information into a structure-controlled mode, to adaptively adjust the imbalance distribution, to reduce the redundant interferences between data sources, and to enhance the cooperative performance. The network proposed by Zhang et al. [44] was embedded by multibranch features to help the fine-grained serialization of multisource features; subsequently, a locally based multisource feature interactor (L-MSFI) was designed for simultaneous exploration of HSI and LiDAR features. It provides an information transfer environment for multibranch features and further alleviates the homogeneous processing model of the self-attentive process.

As mentioned above, some preliminary multimodal networks have been proposed for RS images and obtained better classification results than using only a single modality [45]. However, it is worth noting that the fusion strategy is a critical factor in determining the performance of multimodal networks. Currently, many obstacles in the image fusion process still lead to the loss of critical information, and the fusion strategies proposed so far can only discriminate features having different levels of information [46]. In [47], a pyramid feature attention fusion strategy (PFAF-Net) method based on multiscale features is proposed. Different fusion techniques fuse high and low level features; finally, global and local features are merged effectively. It not only integrates multimodal data but also has rich content details to get better image fusion. In [48], the advantage of pixel-level fusion is fully utilized to avoid information loss that may occur during feature extraction and improve the accuracy of CNN classifiers. Lu et al. [49] argued that remote sensing relies on pixel-level feature fusion classification methods that rely on cascaded or weighted fusion due to different imaging mechanisms, and thus, they considered it crucial to extract and fuse similar higher-order semantic information and complementary discriminative information contained in multimodal data, which led to the proposal of a novel coupled adversarial learning-based classification (CALC) approach. Ding et al. [50] proposed a global–local fusion classification strategy that integrates multiclass estimation based on local and global features through a probabilistic decision fusion strategy. Zhang et al. [51] designed a novel three-branch CNN to learn spectral, spatial, and elevation features, each of which employs a multilevel feature fusion (MLF) module to fuse shallow and deep features. Meanwhile, the information flow between spatial and elevation branches was increased by the mutual guidance attention (MGA) module to fully fuse spatial and elevation information.

In most of the above works, the feature extractor is constructed based on CNNs. However, we deeply mine the spectral features of HSI through the involution operator to improve the relevance of the spectral dimensions. Then, we design a multiscale hierarchical fusion module (MHFM), which adopts a multibranching approach to extract the multiscale information and then forms the spectral–spatial features with a hierarchical fusion strategy that enhances the spectral–spatial information. Meanwhile, the Lidar data is extracted with three-layer convolution to extract shallow features as supplementary information. Finally, a practical adaptive feature fusion module (AFFM) is proposed to fuse and classify the data based on the combined strategy of feature level and weighted fusion, using adaptive weights based on the different contribution degrees. The weights are dynamically assigned to find the optimal model with the constraint of the joint loss of the three. The main contributions of this article are summarized as follows.

1) In this article, deep mining of spectral features of HSI using the involution operator is proposed. The involution operator is a learnable alternative to convolution that captures local features more efficiently than traditional convolution operations. By introducing the involution operator, the model can enhance the correlation of spectral dimensions and improve the performance and efficiency of the feature extractor.
2) Second, in order to further extract hyperspectral image features, this article designs a multiscale hierarchical fusion module (MHFM). MHFM uses a multibranching approach to extract multiscale information and a hierarchical fusion strategy to form spectral–spatial features. By extracting and fusing features at different scales, the MHFM can enhance the expressive ability of spectral–spatial information, which makes the model understand the features of HSI data more comprehensively.
3) Finally, in order to better fuse different features and optimize the classification effect, this article proposes AFFM. The AFFM fuses the data for classification based on the combination strategy of feature level and weighted fusion, then adopts the adaptive weights to assign the contribution degree of different fusion features dynamically. According to the different contributions of the three fusion features obtained from the maximum, adaptive addition, and concat, AFFM can assign corresponding weights based on the contributions. In this way,

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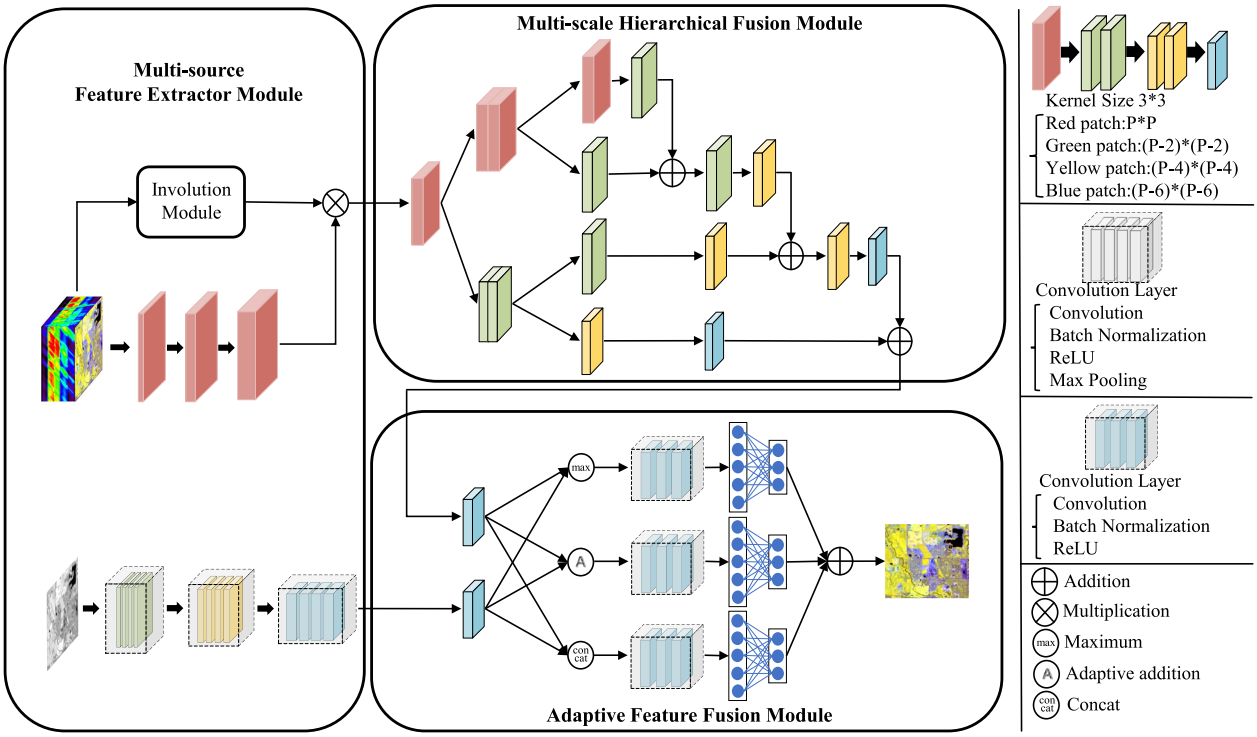


Fig. 1. Illustration of the proposed AMSSE-Net.

it accelerates the convergence of the loss function of the network model, finds the optimal model, and improves the classification performance.

The method in this article achieves significant improvements and superior performance in HSI data processing and fusion of multimodal data. More detailed information about the proposed method is given in Section II. The rest of the article is organized as follows. Section III presents the experimental results and analyses in detail. Finally, Section IV summarizes some concluding remarks.

II. Proposed Approach

This section describes the proposed AMSSE-Net model in detail. Firstly, the general structure of the network is shown in Fig. 1.

For a given hyperspectral image $X_{\mathrm{HSI}} \in R^{M\times N\times D}$ covering the same area of the Earth's surface and the corresponding LiDAR image $X_{\mathrm{LiDAR}} \in R^{M\times N\times d}$. Here, M and N denote the height and width of the remote sensing image, respectively. D and d refer to the number of spectral bands of the hyperspectral image and the number of spectral bands of the LiDAR image, respectively. Since the involution will group the channels of the hyperspectral input data, in order to reduce the redundant spectral information, principal component analysis (PCA) is used to extract the first 15 principal components of the HSI spectral channel. At the same time, the input data is normalized, so it is represented as $X_{\mathrm{HSI}} \in R^{M\times N\times 15}$ and $X_{\mathrm{LiDAR}} \in R^{M\times N\times 1}$.

A. Multisource Data Feature Extractor Module

It is well known that CNN possesses the properties of local sensing, parameter sharing, deep structure, and nonlinear activation function. By employing the principles of local sensing and parameter sharing, this approach adeptly captures the local features of the data. Its deep structure and hierarchical representation facilitate the gradual extraction of higher-level features that exhibit stability and invariance. It is particularly well-suited for processing nonlinear datasets such as hyperspectral and LiDAR data. Therefore, we use CNN to extract hyperspectral and Lidar data features.

Firstly, for hyperspectral data, the features are extracted here mainly to improve the spectral dimensionality correlation by using involution later on and, at the same time, to enable the neurons to capture texture, spatial structure, and multiscale features in the input data in a more flexible way. The hyperspectral feature extractor we designed consists of three convolutional layers for extracting spectral–spatial features from the preprocessed data input X_{hsi}, where $X_{\mathrm{hsi}} \in R^{p\times p\times 15}$, and p denotes the scale of the data block. The three-layer convolutional network contains only three layers of filters without batch normalization (BN) and corrective linear unit (ReLU) activation functions. Specifically, the three layers contain 16, 32, and 64 filters, respectively, to obtain the basic feature f_{hsi}, $f_{\mathrm{hsi}} \in R^{P\times P\times C}$, where C is 64. This implies that the network does not carry out the normalization and nonlinear transformation operations during the feature extraction process, which results in the network being more inclined to retain some features of the original data. Theoretically, it is more reasonable to merge the data without additional transformations and processing with the data obtained by the involution operator, which is helpful for deeper mining of spectral features.

Inspired by [39] and [52], we introduce the involution operator to mine the spectral information deeply. The involution operator is generated as shown in Fig. 2. The involution operator has symmetric inverse properties, that is, spatial specificity

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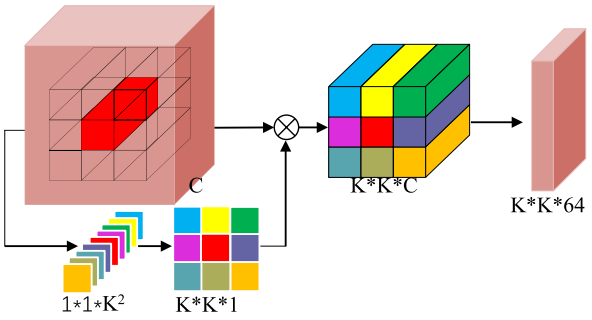


Fig. 2. Involution module. The input channels are divided into groups of G (with G equal to 1 as an example), and the consequent generation of the involution kernel is of size $1 \times 1 \times K^2$, which is rearranged to $K \times K \times 1$. Next, the involution kernel is multiplied with the input data. Finally, a layer of Conv2d is designed to get $K \times K \times 64$ features.

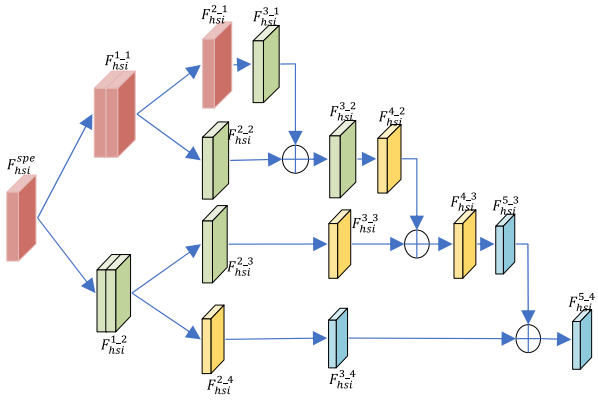


Fig. 3. Multiscale hierarchical fusion module.

and channel invariance, compared to the regular convolution. For the pixel $X_{\text{hsi}}^{i,j}$ at position (i, j), where $X_{\text{hsi}}^{i,j} \in R^{1\times1\times15}$, firstly, the channels are grouped, and the 15 spectral channels are divided into G groups, each group having its specific pairwise kernel. At each different position, the involution operator has a different kernel, so the kernel of involution is automatically generated based on the input features, which are feature-mapped to form a dynamic convolution kernel, ensuring that the kernel size and the input feature size can be automatically aligned in the spatial dimension. Specifically, for the input data $X_{\text{hsi}}^{i,j}$, a pairwise kernel with kernel size $K\times K$ is defined as $H_g \in R^{K\times K}$, $g = 1, 2, \ldots,$ G. G calculates the number of clusters sharing the same pairwise kernel within the cluster. Then, a channel-to-space rearrangement is created. The channel shapes can be rearranged from a vector $1 \times 1 \times K^2 \times G$ to a tensor $K \times K \times 1 \times G$. Finally, a multiply-add operation is performed. The pairwise kernel H is multiplied by the input tensor of all channels in the same group to join the G groups together to obtain a $K \times K \times 15$ tensor, which is summed over a $K \times K$ spatial neighborhood and aggregated over all spatial neighborhoods. The specific formula can be expressed as

$$F_{\text{hsi}}^{i,j,g} = \sum_{(u,v)\in\Delta_K} H_{i,j,u+\lfloor K/2\rfloor,v+\lfloor K/2\rfloor,g} X_{\text{hsi}}^{i+u,j+v,g} \tag{1}$$

where $\Delta_K = [-\lfloor K/2\rfloor, \ldots, \lfloor K/2\rfloor] \times [-\lfloor K/2\rfloor, \ldots, \lfloor K/2\rfloor]$ refers to a set of offsets in the neighborhood of the center pixel, and the symbol $\times$ denotes the Cartesian product. Finally, dimension matching is achieved by 1×1 convolution. The feature $F_{\text{hsi}}^{i,j}$ obtained from the involution operator is multiplied by the basic feature f_{hsi} to obtain the spectral feature. Finally, we get the spectral feature representation of the HSI, $F_{\text{hsi}}^{\text{spe}}$, $F_{\text{hsi}}^{\text{spe}} \in R^{P\times P\times C}$.

In addition, the feature extractor for LiDAR images designed in this article is mainly used to compensate for the loss of elevation information in pixel fusion. Since LiDAR images are single-channel images, the deeper convolutional layers can capture more abstract and semantic features. Still, they are less valuable than the shallow network for extracting our desired elevation features. Therefore, we use a three-layer shallow network to learn the elevation features, specifically, the kernel size of all the convolutional layers is set to 3×3, and each convolutional layer is followed by a batch normalization layer, a ReLU layer, and a max-pooling layer, which ultimately results in the LiDAR's feature representation F_{LiDAR}, which is $F_{\text{LiDAR}} \in R^{(P-6)\times(P-6)\times C}$.

B. Multiscale Hierarchical Fusion Module

In order to extract multiscale information from the data, the multiscale hierarchical fusion (MHF) module is designed, as shown in Fig. 3. The MHF module consists of three splitting operations and three fusion operations.

Firstly, the first splitting operation splits the extracted spectral features $F_{\text{hsi}}^{\text{spe}}$ into two groups. Specifically, $F_{\text{hsi}}^{\text{spe}}$ is split into two branches after parallel CNN. The number of channels is enlarged to twice that of the original one, which is $F_{\text{hsi}}^{1_1} \in R^{P\times P\times 2C}$, and $F_{\text{hsi}}^{1_2} \in R^{(P-2)\times(P-2)\times 2C}$, respectively. Thus, the first splitting operation is defined as

$$F_{\text{hsi}}^{1_1}, F_{\text{hsi}}^{1_2} = S_1\left(F_{\text{hsi}}^{\text{spe}}\right) \tag{2}$$

where S_i denotes the ith splitting operation. The second and third splitting operations are also divided into two branches after parallel CNN. Still, the number of channels is reduced to half of the original one, which is consistent with the number of F_{hsi} channels, and is denoted as $F_{\text{hsi}}^{2_1} \in R^{P\times P\times C}$, $F_{\text{hsi}}^{2_2} \in R^{(P-2)\times(P-2)\times C}$, $F_{\text{hsi}}^{2_3} \in R^{(P-2)\times(P-2)\times C}$, $F_{\text{hsi}}^{2_4} \in R^{(P-4)\times(P-4)\times C}$. The second and third splitting operations are defined as

$$F_{\text{hsi}}^{2_1}, F_{\text{hsi}}^{2_2} = S_2\left(F_{\text{hsi}}^{1_1}\right) \tag{3}$$

$$F_{\text{hsi}}^{2_3}, F_{\text{hsi}}^{2_4} = S_3\left(F_{\text{hsi}}^{1_2}\right). \tag{4}$$

Then, the dimensions of the branches are reduced by a 3×3 convolutional layer for $F_{\text{hsi}}^{2_3}$ and $F_{\text{hsi}}^{2_4}$ while increasing the sensory field of the branch in which they are located, and the feature extraction and representation learning are performed for each branch, which is denoted as $F_{\text{hsi}}^{3_3} \in R^{(P-4)\times(P-4)\times C}$, $F_{\text{hsi}}^{3_4} \in R^{(P-6)\times(P-6)\times C}$. Up to this point, multiscale feature extraction has been achieved through multilayer convolutional operations, $F_{\text{hsi}}^{2_1}$, $F_{\text{hsi}}^{2_2}$, $F_{\text{hsi}}^{3_3}$, and $F_{\text{hsi}}^{3_4}$, respectively. The feature maps with smaller receptive fields can better capture the detailed information. In contrast, the feature maps with larger receptive fields can capture the more macroscopic structural and semantic information, which makes the network more robust and generalized when dealing with different data.

In order to integrate the extracted multiscale information better, a layered fusion strategy is used to form a richer and integrated feature representation. Firstly, for the feature map

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$F_{\text{hsi}}^{2_1}$ with a small sensory field, a 3×3 convolutional layer is first passed to make the feature map match the dimension of $F_{\text{hsi}}^{2_2}$, which is denoted as $F_{\text{hsi}}^{3_1} \in R^{(P-2)\times(P-2)\times C}$, and then $F_{\text{hsi}}^{3_1}$ and $F_{\text{hsi}}^{2_2}$ are superimposed. The first layer fusion feature map $F_{\text{hsi}}^{3_2}$ is obtained, denoted as $F_{\text{hsi}}^{3_2} \in R^{(P-2)\times(P-2)\times C}$. Meanwhile, the first layer feature fusion operation is defined as

$$F_{\text{hsi}}^{3_1} = C_1\left(F_{\text{hsi}}^{2_1}\right) \tag{5}$$

$$F_{\text{hsi}}^{3_2} = F_{\text{hsi}}^{3_1} + F_{\text{hsi}}^{2_2} \tag{6}$$

where C_j denotes the convolution operation of the jth layer. The feature fusion operations in the second and third layers are similar to those in the first layer, both by superimposing feature maps from different perceptual domains from different branches, which are defined as

$$F_{\text{hsi}}^{4_2} = C_2\left(F_{\text{hsi}}^{3_2}\right) \tag{7}$$

$$F_{\text{hsi}}^{4_3} = F_{\text{hsi}}^{4_2} + F_{\text{hsi}}^{3_3} \tag{8}$$

$$F_{\text{hsi}}^{5_3} = C_3\left(F_{\text{hsi}}^{4_3}\right) \tag{9}$$

$$F_{\text{hsi}}^{5_4} = F_{\text{hsi}}^{5_3} + F_{\text{hsi}}^{3_4}. \tag{10}$$

The multiscale fusion feature $F_{\text{hsi}}^{5_4}$, denoted as $F_{\text{hsi}}^{5_4} \in R^{(P-6)\times(P-6)\times C}$, is obtained through three layers of fusion, followed by the BN layer for regularization and acceleration of the training process and the corrective linear unit (ReLU) for learning the nonlinear representation, to obtain the final feature map F_{hsi}, $F_{\text{hsi}} \in R^{(P-6)\times(P-6)\times C}$. This helps the model to capture both detail and global context information. In addition, the strategy of hierarchical fusion compensates for the information loss to some extent and improves the feature delivery.

C. Adaptive Feature Fusion Module

After obtaining the feature representation F_{hsi} for hyperspectral and F_{LiDAR} for LiDAR, in order to make full use of the spatial–spectral information of the hyperspectral and the elevation information of the Lidar data, the data is fused using a combined strategy based on feature level and weighted fusion. In feature fusion, common methods include maximum, adaptive addition, and concat.

1) Maximum: In this fusion method, the feature with the maximum value is selected as the final fusion result for features at the same location. This method extracts the most significant and salient information from the multimodal data features and helps emphasize the image's most important feature parts. This fusion method is expressed as

$$F_1 = \text{Maximum}(F_{\text{hsi}}, F_{\text{LiDAR}}). \tag{11}$$

2) Adaptive Addition: In adaptive addition fusion, features at the same location are summed element by element. This fusion method accumulates the information between different features and enhances the feature representation. Additive fusion helps to combine information from multiple features to provide a more comprehensive and enriched feature representation. However, considering that different features contribute to the final classification results in different degrees in the classification task. For hyperspectral and LIDAR data, hyperspectral data focuses on spectral–spatial information. LiDAR data is mainly used to make up for the missing elevation information of hyperspectral data, so in this article, we adopt adaptive weighting α to dynamically assign weights, that is, weighted fusion, according to the different attributes and the degree of contribution of the data. Also, this helps to balance the pixel values and facilitates the integration of the three fusion features. So, this fusion method is expressed as

$$F_2 = \alpha_1 \times F_{hsi} + \alpha_2 \times F_{\text{Lidar}}, \quad \alpha_1 + \alpha_2 = 1. \tag{12}$$

3) Concat: In stitching fusion, different features are spliced according to the channel dimension. This approach preserves the integrity of information between individual features while taking into account the interrelationships and interactions between hyperspectral and Lidar features. This fusion method is expressed as

$$F_3 = \text{Concat}(F_{\text{hsi}}, F_{\text{Lidar}}). \tag{13}$$

The three obtained features are passed through the convolutional, batch normalization, and ReLU layers. In particular, for splicing fusion, the channel dimension of the fused data becomes double of the original, so a 3×3 convolutional layer is added to restore it to the original number of channels. Finally, after a fully connected layer, we obtain $O_1 \in R^{N\times 1}$, $O_2 \in R^{N\times 1}$, $O_3 \in R^{N\times 1}$, respectively, where N denotes the number of categories to be distinguished. Thus, the final output O is defined as

$$O = O_1 + O_2 + O_3. \tag{14}$$

For O_1, O_2, and O_3, their loss values are computed by means of the cross-entropy loss function. The cross-entropy loss function is expressed as

$$\text{Loss}_k = -\frac{1}{\text{num}} \sum_{i} \sum_{j=1}^{N} y_{ij} \log O_k^{ij}, \quad k = 1, 2, 3 \tag{15}$$

where num denotes the number of samples, N denotes the number of categories to be distinguished, and y_{ij} denotes the sign function, which takes 1 when the true category of sample i is equal to j, and 0 otherwise. O_k^{ij} denotes the probability that sample i is predicted to be j. Because the three fusion data do not have the same focus, adaptive weight β is used to find the optimal model by dynamically assigning the weights according to the different contributions of the three outputs. The final loss value L is expressed as

$$L = \beta_1 \times \text{Loss}_1 + \beta_2 \times \text{Loss}_2 + \beta_3 \times \text{Loss}_3. \tag{16}$$

Therefore, the AFFM is used to fuse features from hyperspectral and LiDAR data to improve classification performance. The module involves three different feature fusion methods: maximum, adaptive addition, and concat. The three fusion methods integrate information from multiple sources of data to improve the comprehensive feature representation. Among them, adaptive addition fusion also introduces weights (α_1 and α_2) to dynamically assign weights to reflect the contribution of different features to the final classification result. α_1 and α_2 add up to 1, balancing the pixel values of different

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TABLE I
NUMBER OF SAMPLES IN EACH CLASS OF THE TRENTO DATASET

Class No.	Class Name	Number of Samples Train	Test
1	Apple Trees	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3052
-	Total	819	29395

feature fusion methods. This balancing concept not only ensures that the data from different sources interact with each other in an appropriate manner, but also equalizes the pixel weights of the three fusion methods. The three feature fusion methods were used to obtain outputs O_1, O_2, and O_3. For each output (O_1, O_2, and O_3), loss values (Loss_1, Loss_2, and Loss_3) were computed using the cross-entropy loss function. These loss values measure the classification performance of the respective fused features. To handle the different concerns of different fusion methods, adaptive weights β_1, β_2, and β_3 are introduced. These weights are used to compute the final loss value L to combine the losses of the three outputs, assigning the appropriate weights based on the contributions. In this way, it accelerates the convergence of the loss function of the network model, finds the optimal model, and improves the classification performance.

III. EXPERIMENTS

Three widely used remote sensing datasets are used to evaluate the proposed AMSSE-Net framework. All procedures were performed using an Intel[1] Core[2] i5-8300H CPU and an Nvidia 1050Ti GPU.

We implemented AMSSE-Net using the PyTorch framework. We trained 200 epochs on a single GPU with the initial learning rate set to 0.01, and the learning rate was updated every 80 epochs with a drop of 1/2. During the training phase, Adam was used as an optimizer.

A. Data Description

1) Trento data was collected in a rural area south of Trento, Italy. The data consists of 600 × 166 pixels with a spatial resolution of 1 m. The HSI data captured by the AISA Eagle sensor consists of 63 spectral bands covering from 0.42 to 0.99 μm. The LiDAR data was acquired by the Optech ALTM 3100EA sensor. Table I lists class names, number of training, and test samples.
2) MUUFL data was acquired on 8 November 2010 at the University of Southern Mississippi, Gulf Park campus. The data collection consisted of hyperspectral and LiDAR data for the corresponding areas of the campus. The HSI data acquired by the CASI-1500 imaging sensor consisted of 64 spectral bands ranging from 375 to 1050 nm with a spatial resolution of 0.54 m. The data were collected by the CASI-1500 imaging sensor. Table II lists class names, number of training, and test samples.
3) Houston 2013 data was acquired by the ITRES CASI-1500 imaging sensor on the University of Houston campus and surrounding rural areas in Houston, TX, USA, courtesy of the 2013 IEEE GRSS Data Fusion Competition. The HSI data consisted of 349 × 1905 pixels containing 144 bands in the spectral range from 364 to 1046 nm with a spectral resolution of 10 nm, while only one band of LiDAR data provides elevation information for the same image area. Table III lists class names, number of training, and test samples.

[1]Registered trademark.
[2]Trademarked.

TABLE II
NUMBER OF SAMPLES IN EACH CLASS OF THE MUUFL DATASET

Class No.	Class Name	Number of Samples Train	Test
1	Trees	150	23096
2	Mostly grass	150	4120
3	Mixed ground surface	150	6732
4	Dirt and sand	150	1676
5	Road	150	6537
6	Water	150	316
7	Building Shadow	150	2083
8	Building	150	6090
9	Sidewalk	150	1235
10	Yellow curb	150	33
11	Cloth panels	150	119
-	Total	1650	52037

TABLE III
NUMBER OF SAMPLES IN EACH CLASS OF THE HOUSTON DATASET

Class No.	Class Name	Number of Samples Train	Test
1	Health grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
-	Total	2832	12197

B. Parameter Adjustments

Effect of batch and patch scale size: We studied the impact of several different-sized data blocks under different batch sizes. The performance of different scale sizes (7 × 7, 9 × 9, 11 × 11, 13 × 13, 15 × 15, and 17 × 17) under different batch sizes (16, 32, 64, and 128) is shown in Tables IV–VI. On the Trento data, the overall accuracy (OA) values are generally higher than the other batch sizes when the batch size is 16, and the highest OA values are found when the scale size is 17. On the MUUFL data, at any batch size, OA values peaked when the scale size was 13; OA reached a maximum when the batch size was 128. On the Houston data, OA improves by more than 1% overall compared to other scales when the

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TABLE IV

CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD WITH DIFFERENT BATCH SIZES AND DIFFERENT PATCH SCALES ON THE TRENTO DATASET

batch size	metrics	patch size 7	9	11	13	15	17
16	OA	97.25	97.55	96.58	98.56	97.86	**98.79**
	AA	95.72	94.58	92.53	95.75	94.48	**95.49**
	Kappa	0.9632	0.9671	0.9544	0.9807	0.9713	**0.9838**
32	OA	97.99	96.87	97.19	97.67	98.16	97.40
	AA	96.85	95.74	96.29	94.76	94.68	93.78
	Kappa	0.9730	0.9583	0.9625	0.9688	0.9753	0.9653
64	OA	96.57	94.34	94.26	98.44	95.92	95.41
	AA	95.50	93.52	94.60	96.10	96.03	94.11
	Kappa	0.9543	0.9250	0.9241	0.9790	0.9458	0.9391
128	OA	92.41	96.76	94.56	94.25	82.38	97.40
	AA	92.61	94.40	94.82	93.15	85.91	95.55
	Kappa	0.8997	0.9569	0.9279	0.9240	0.7746	0.9653

TABLE V

CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD WITH DIFFERENT BATCH SIZES AND DIFFERENT PATCH SCALES ON THE MUUFL DATASET

batch size	metrics	patch size 7	9	11	13	15	17
16	OA	87.65	89.58	89.79	89.22	89.30	84.05
	AA	88.06	91.20	91.69	91.02	90.80	86.86
	Kappa	0.8379	0.8637	0.8664	0.8584	0.8598	0.7936
32	OA	87.82	89.43	88.31	90.07	88.84	87.64
	AA	89.58	91.68	90.88	91.41	90.89	90.40
	Kappa	0.8410	0.8619	0.8480	0.8697	0.8541	0.8390
64	OA	87.24	88.11	90.17	90.82	90.51	90.34
	AA	89.05	90.31	91.93	92.64	91.73	92.23
	Kappa	0.8340	0.8450	0.8715	0.8796	0.8754	0.8734
128	OA	87.53	90.07	90.47	**91.48**	90.55	90.26
	AA	89.93	91.65	91.91	**91.87**	91.99	92.07
	Kappa	0.8378	0.8699	0.8749	**0.8875**	0.8761	0.8723

TABLE VI

CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD WITH DIFFERENT BATCH SIZES AND DIFFERENT PATCH SCALES ON THE HOUSTON DATASET

batch size	metrics	patch size 7	9	11	13	15	17
16	OA	84.27	90.41	89.73	88.05	88.38	88.09
	AA	86.01	91.57	90.50	88.43	89.01	88.74
	Kappa	0.8291	0.8961	0.8885	0.8703	0.8738	0.8707
32	OA	86.96	88.21	**92.26**	89.37	91.35	89.74
	AA	89.11	89.31	**92.10**	90.52	92.33	89.70
	Kappa	0.8586	0.8719	**0.9159**	0.8848	0.9061	0.8886
64	OA	87.97	86.94	91.43	88.48	89.33	88.23
	AA	89.28	88.76	90.92	89.46	89.79	88.95
	Kappa	0.8694	0.8589	0.9069	0.8751	0.8843	0.8725
128	OA	86.55	87.77	89.96	86.87	90.36	88.51
	AA	88.31	89.72	90.89	88.17	90.68	87.42
	Kappa	0.8540	0.8674	0.8912	0.8579	0.8955	0.8751

scales are at 9 and 11; OA peaks at a batch size of 32 and a scale size of 11.

C. Ablation Studies and Module Analysis

In order to validate the proposed method, we conducted a series of ablation experiments to assess its impact on classification performance by comparing the classification results of different structural configurations. Through the ablation experiments, we can identify key network modules and design decisions to optimize the design of the neural network further. This helps us better understand how DL networks work, improve classification performance, and provide valuable guidance for future research and applications.

TABLE VII

CLASSIFICATION PERFORMANCE OF DIFFERENT COMBINATIONS OF IM, MHFM, AND AFFM ON TRENTO, MUUFL, AND HOUSTON DATASETS

Module: Involution Module	Module: Multi-scale Hierarchical Fusion Module	Module: Adaptive Feature Fusion Module	Metrics	Datasets: Trento	MUUFL	Houston
✓	✗	✗	OA	92.83	90.38	79.04
			AA	76.16	92.35	80.84
			Kappa	0.9042	0.8741	0.7723
✗	✓	✗	OA	95.65	89.28	90.22
			AA	91.35	89.67	90.43
			Kappa	0.9418	0.8838	0.8937
✗	✗	✓	OA	95.60	89.94	87.57
			AA	91.32	91.75	88.64
			Kappa	0.9413	0.8683	0.8653
✓	✓	✗	OA	97.68	90.71	90.69
			AA	93.40	92.20	91.56
			Kappa	0.9690	0.8780	0.8989
✗	✓	✓	OA	94.75	91.22	90.68
			AA	89.11	92.74	91.23
			Kappa	0.9296	0.8845	0.8989
✓	✗	✓	OA	93.87	90.12	90.63
			AA	90.42	91.78	90.81
			Kappa	0.9181	0.8706	0.8984
✓	✓	✓	OA	**98.79**	**91.48**	**92.26**
			AA	**95.49**	**91.87**	**92.10**
			Kappa	**0.9838**	**0.8875**	**0.9159**

1) Effect of Involution Module, MHFM, and AFFM: In our proposed network, involution module (IM) and MHFMs are the modules for mining deep spectral–spatial features of HSI. The AFFM is the multisource feature fusion module in our proposed network, and the three of them synergize to improve the classification performance of the network. Therefore, in order to verify the effectiveness of IM, MHFM, and AFFM, we perform ablation experiments on the three modules. The feature extraction module IM and the MHFM both use one layer of conv2d for the transition in terms of the number of channels and data block size matching, and the replacement multisource feature fusion module AFFM uses simple summation for fusion. We test the final classification results on three datasets using the same experimental setup. The experimental results for the last module configuration are shown in Table VII. From the results in the table, it can be inferred that the combination of IM+MHFM+AFFM achieves the best classification results on all datasets.

Specifically, the use of IM compared to the use of IM+MHFM and IM+AFFM reveals that in the Trento dataset, the OA is improved by 4.85% and 1.04%, respectively, but the average accuracy (AA) is significantly improved by more than 10%. In the Houston dataset, OA is improved by 11.65% and 11.59%, respectively, which confirms to some extent the ability of MHFM to mine multiscale spatial information and the effectiveness of AFFM for multisource feature fusion.

A comparison of using MHFM to using IM+MHFM and MHFM+AFFM reveals that there is not much difference in the classification performance between using MHFM and using MHFM+AFFM in the Trento dataset. Still, the OA is improved by at least 2% for IM+MHFM, which suggests that the combination of IM and MHFM during the feature extraction step can be more effective in improving the capture of spatial and spectral features. In the MUUFL dataset, the combination of IM+MHFM and MHFM+AFFM improved OA by 1.43% and 1.94%, respectively, over MHFM alone.

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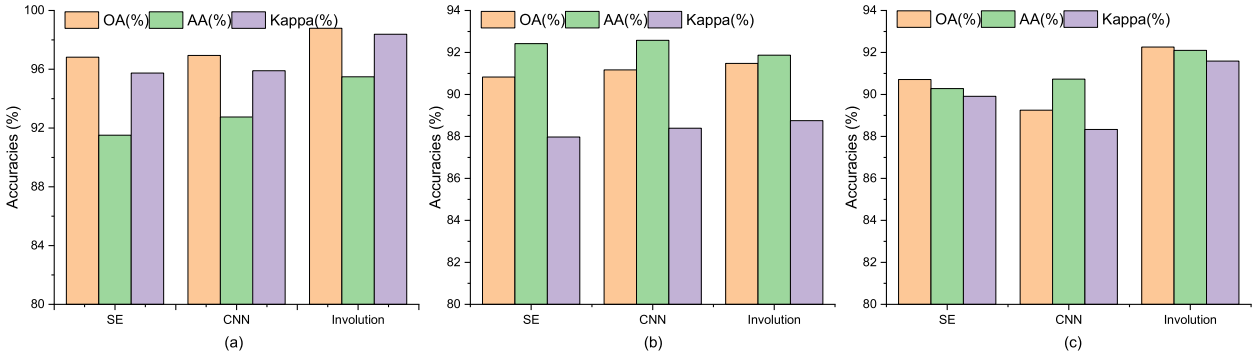


Fig. 4. Spectral feature extraction efficacy studies results in terms of OA (%), AA (%), and Kappa ×100 (%) about SE block, CNN, and involution module. (a) Trento dataset. (b) MUUFL dataset. (c) Houston dataset.

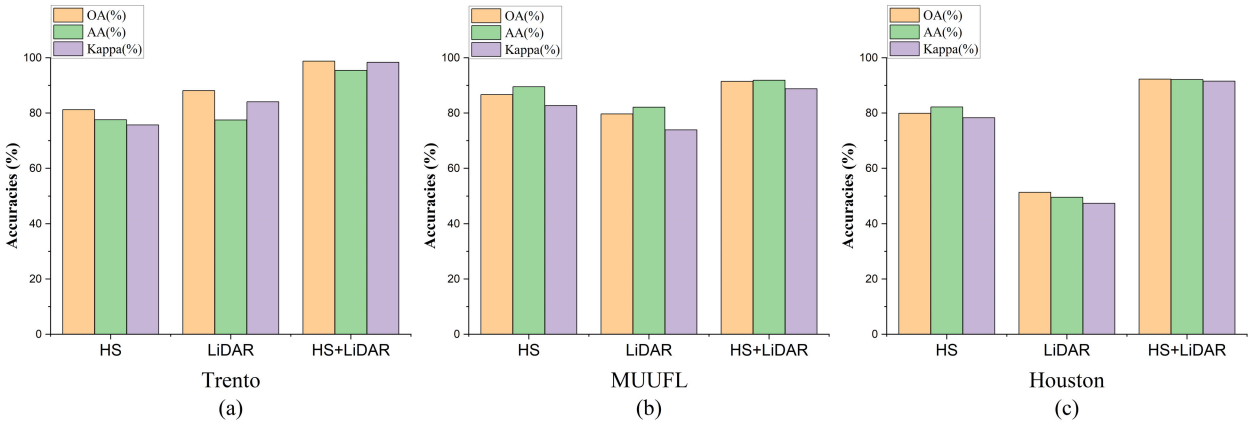


Fig. 5. Ablation study results in terms of OA (%), AA (%), and Kappa ×100 (%) about the data source usage (i.e., only HS, only LiDAR, and both HS and LiDAR). (a) Trento dataset. (b) MUUFL dataset. (c) Houston dataset.

In the Houston dataset, the other two improved OA slightly over MHFM alone, but AA improved by about 1%.

Comparison of the use of AFFM using IM+AFFM and MHFM+AFFM reveals that in the Trento dataset, although the OA is slightly better using AFFM alone, this is partly a reflection of the importance of combining IM and MHFM to mine spatial and spectral features jointly. In the MUUFL and Houston datasets, the use of IM+AFFM and MHFM+AFFM outperforms AFFM alone in terms of OA, AA, and Kappa. The combination of IM+MHFM+AFFM achieves the best classification results on all datasets.

2) Analysis of the Effectiveness of Involution in Spectral Feature Extraction: To illustrate the effectiveness of spectral feature extraction by the involution module, we compare the classification performance of the involution module, squeeze and excitation (SE) block [53], and CNN in the network, respectively. Fig. 4 shows the detailed comparison results of the involution module, SE block, and CNN in terms of OA, AA, and Kappa coefficients.

The classification performance of the three spectral feature extraction methods on three different datasets can be observed in Fig. 4. The data results show that the involution module achieves the optimal results in terms of OA and Kappa on all three datasets, indicating that the model has a better performance in general. In particular, AA on the MUUFL dataset is slightly underperforming, with a performance difference of about 0.7% from the other two methods. This suggests that the model may be performing well in the large categories, but mediocre on the small categories, resulting in a lower AA. On the Houston and Trento datasets, the use of the involution module to extract spectral information achieved better results in terms of OA, AA, and Kappa compared to SE block and CNN. Specifically, the OA, AA, and Kappa improvement on the Trento dataset using the involution module is significant, with an overall accuracy OA improvement of about 2% on the whole dataset. On the Houston dataset, the OA improvement is about 1.5% compared to the suboptimal SE block and about 3% compared to CNN.

3) Fusion Method and Adaptive Parameter Analysis: In order to further validate the superiority of the AFFM while reflecting the superiority of multimodal data fusion for classification, we first compared the performance of unimodal and multimodal classification. Specifically, we first trained the two branches of the model, HSI and LiDAR, separately and compared the classification performance of each of the two branches. Then, we verify the effectiveness of the adaptive weighting mechanism by comparing the loss curves of the three fusion methods and the adaptive loss curve. Finally, the superiority of the classification performance of combining the three fusion methods is illustrated based on different combinations of the three fusion methods.

Fig. 5 shows the detailed comparison results between unimodal and multimodal data in terms of OA, AA, and Kappa coefficients. It is clear that multimodal data classification shows significant performance advantages in terms of OA, AA, and Kappa. Figs. 6–8 show the loss curves of the

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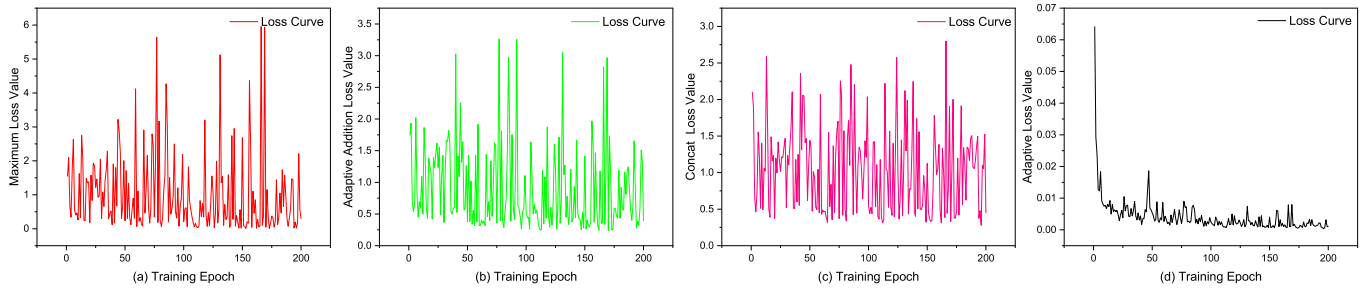


Fig. 6. Loss curves of the three fusion methods and the adaptive loss curve in the Trento dataset. (a) Maximum loss curve. (b) Adaptive addition loss curve. (c) Concat loss curve. (d) Adaptive loss curve.

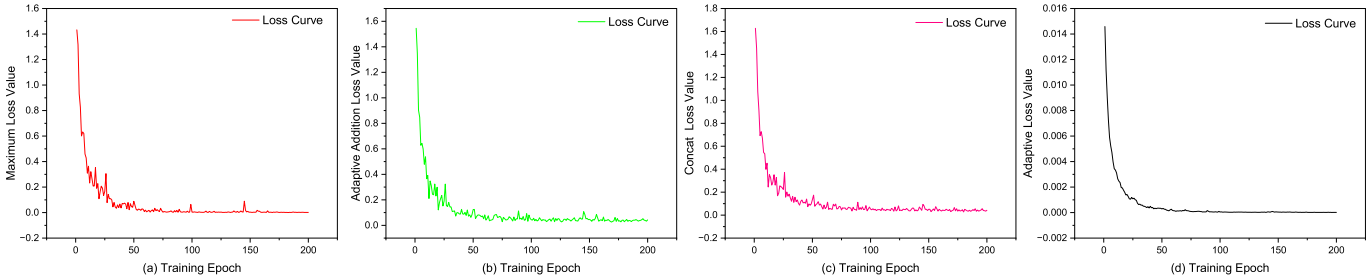


Fig. 7. Loss curves of the three fusion methods and the adaptive loss curve in the MUUFL dataset. (a) Maximum loss curve. (b) Adaptive addition loss curve. (c) Concat loss curve. (d) Adaptive loss curve.

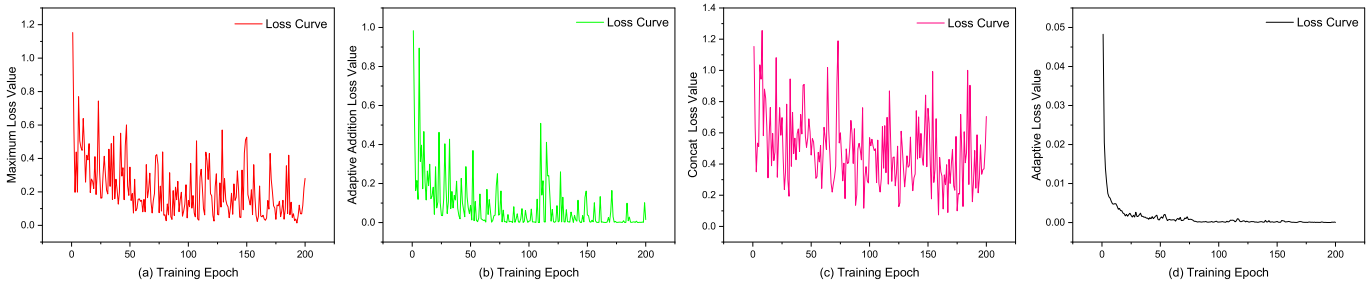


Fig. 8. Loss curves of the three fusion methods and the adaptive loss curve in the Houston dataset. (a) Maximum loss curve. (b) Adaptive addition loss curve. (c) Concat loss curve. (d) Adaptive loss curve.

three fusion methods and the adaptive loss curves on the three datasets. The loss curves for each of the three fusion methods in Figs. 6 and 8 show large fluctuations. Noise exists in the data itself, and in the process of feature fusion of multisource data, more noise is inevitably introduced, while causing information redundancy. For the noise present in the data itself, we assign adaptive weights to the HSI and Lidar features to reduce the impact of noise-sensitive features. From Fig. 8(b), we can find that the adaptive addition loss curve is the most stable and less fluctuating compared to the loss curve of Fig. 8(a) and (c), which shows the effectiveness of adaptive addition. Of course, the final adaptive loss curve performs best on all three datasets, and its loss curve converges quickly and fluctuates less. It proves that the adaptive addition mechanism can improve the model performance and accelerate the finding of the optimal model.

The loss function of the model proposed in (16) contains three hyperparameters (i.e., β_1, β_2, and β_3). Since β_1, β_2, and β_3 are multiplied by the cross-entropy losses obtained from the outputs of the three different fusion methods (i.e., maximum, adaptive addition, and concat), respectively. We, therefore, test the adaptive weighting mechanism by comparing the loss curves of the three fusion methods with the adaptive loss curves alone to illustrate the effect of the adaptive weights in the loss function on the classification performance.

TABLE VIII

CLASSIFICATION PERFORMANCE OF DIFFERENT COMBINATIONS OF FUSION METHODS OF THE ADAPTIVE FUSION MODULE ON TRENTO, MUUFL, AND HOUSTON DATASETS

Adaptive Feature Fusion Module			Metrics	Datasets		
Maximum	Adaptive Addition	Concat		Trento	MUUFL	Houston
✓	✗	✗	OA	95.73	89.57	90.01
			AA	92.66	91.69	91.02
			Kappa	0.9431	0.8637	0.8918
✗	✓	✗	OA	96.71	90.03	75.31
			AA	90.76	91.93	79.59
			Kappa	0.9557	0.8694	0.7320
✗	✗	✓	OA	97.79	89.45	91.52
			AA	94.27	91.62	91.64
			Kappa	0.9704	0.8619	0.9079
✓	✓	✗	OA	97.89	90.99	87.28
			AA	93.97	91.54	88.87
			Kappa	0.9718	0.8818	0.8619
✗	✓	✓	OA	89.68	89.46	91.28
			AA	90.30	91.67	92.27
			Kappa	0.8655	0.8624	0.9056
✓	✗	✓	OA	96.86	90.99	89.92
			AA	93.58	91.83	91.35
			Kappa	0.9581	0.8818	0.8906
✓	✓	✓	OA	**98.79**	**91.48**	**92.26**
			AA	**95.49**	**91.87**	**92.10**
			Kappa	**0.9838**	**0.8875**	**0.9159**

Finally, different combinations of these three fusion methods are tested, which can fully reflect the strengths and weaknesses of the fusion strategies. Table VIII reflects the detailed

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TABLE IX
QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND KAPPA ON THE TRENTO DATASET. THE BEST ONE IS SHOWN IN BOLD

No.	Class Name	Performance								
		AM^{3}Net	CALC	CCR-Net	CNN-DF-M	EndNet	GLT-Net	IP-CNN	S^{2}Enet	AMSSE-Net
1	Apple Trees	93.37	98.85	**99.97**	98.75	91.37	99.39	97.08	98.49	99.44
2	Buildings	99.50	**100.00**	99.60	98.74	96.65	99.28	95.36	99.71	99.42
3	Ground	**99.73**	76.20	82.89	98.93	95.72	86.10	99.47	88.50	81.82
4	Woods	98.31	**100.00**	99.99	**100.00**	99.21	**100.00**	97.75	**100.00**	**100.00**
5	Vineyard	98.67	99.84	96.22	99.94	85.50	99.82	95.12	**99.98**	99.81
6	Roads	82.73	90.30	90.56	85.68	89.15	88.50	55.34	90.89	**92.46**
	OA	96.29	98.48	97.43	98.19	92.03	98.41	92.13	98.67	**98.79**
	AA	95.38	94.20	94.87	**97.01**	92.93	95.51	90.02	96.26	95.49
	Kappa	0.9504	0.9796	0.9657	0.9758	0.8944	0.9787	0.8953	0.9822	**0.9838**

TABLE X
QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND KAPPA ON THE MUUFL DATASET. THE BEST ONE IS SHOWN IN BOLD

No.	Class Name	Performance								
		AM^{3}Net	CALC	CCR-Net	CNN-DF-M	EndNet	GLT-Net	IP-CNN	S^{2}Enet	AMSSE-Net
1	Trees	89.41	93.42	88.76	**94.50**	86.15	93.60	88.28	90.77	94.10
2	Mostly grass	68.91	86.65	74.59	**89.30**	82.67	82.72	83.74	76.12	85.15
3	Mixed ground surface	78.21	**86.17**	75.40	78.16	77.18	83.85	81.15	83.60	82.92
4	Dirt and sand	93.97	91.71	90.10	92.18	92.30	89.62	94.33	92.48	**96.30**
5	Road	82.85	86.77	86.25	81.00	**91.86**	88.10	89.12	90.76	91.01
6	Water	**100.00**	99.68	99.37	99.05	98.73	99.05	99.05	**100.00**	99.37
7	Building Shadow	84.64	94.82	93.37	82.91	88.96	95.34	93.04	**95.54**	95.34
8	Building	93.04	95.04	86.93	**98.83**	92.99	95.80	94.68	93.60	94.58
9	Sidewalk	48.58	84.53	76.92	64.94	81.94	84.78	85.10	**90.53**	81.78
10	Yellow curb	69.70	93.94	87.88	90.91	93.94	96.97	93.94	87.88	90.91
11	Cloth panels	95.80	**99.16**	**99.16**	**99.16**	**99.16**	97.48	99.16	**99.16**	**99.16**
	OA	84.99	91.14	85.41	89.58	86.55	90.82	88.25	89.32	**91.48**
	AA	82.28	**91.99**	87.16	88.27	89.63	91.57	91.05	90.95	91.87
	Kappa	0.8034	0.8833	0.8100	0.8618	0.8253	0.8790	0.8471	0.8603	**0.8875**

comparison results of the combinations of different fusion methods in terms of OA, AA, and Kappa coefficient. From the table, it is possible to observe the classification performance of different combinations of the three fusion methods on three different datasets. The data results show that with a single fusion method, the Trento and Houston data achieve the best performance using Concat's fusion method, with an improvement of more than 1% in OA compared to the other fusion methods. In contrast, the MUUFL data achieves the best performance using the adaptive addition method, but the performance difference between it and the other two methods is around 0.5%. MUUFL data achieved the best performance using adaptive addition, but the performance difference with the other two methods was around 0.5%. It can be seen that the single-fusion method using concat is the best, but this does not mean that the other two methods are inferior to concat. In some categories of classification, the other two methods may achieve better results; the advantage of using adaptive parameters is reflected here. Observing the data results, it can be found that the three fusion methods in two-by-two combinations have their advantages on each of the three datasets. At the same time, the overall use of adaptive parameters improves the classification performance when compared to a single-fusion method. Finally, the method using three fusion methods and three hyperparameters performs the best in terms of classification performance with a peak OA and an improvement of more than 1% in OA over the other methods. Thus, AFFM and adaptive parameters can effectively handle features from different sources and achieve more significant improvements in classification accuracy.

D. Comparative Experiments and Analysis

In order to validate the effectiveness of our proposed method, we selected several competitive methods for fusion classification of hyperspectral and LiDAR data to compare with our proposed method, which is evaluated mainly in terms of OA, AA, accuracy per category, and Kappa coefficient. OA defines the ratio between the number of correctly classified pixels and the total number of pixels in the test set. AA is the average of the accuracy across all categories. Kappa is the percentage of consistency corrected by the amount of consistency expected by pure chance.

1) Comparison of Different Classification Methods: In Tables IX–XI, the classification performance of the Trento, MUUFL, and Houston datasets for each model is presented, and the networks used for comparison with our proposed network AMSSE-Net include adaptive mutual-learning-based multimodal data fusion network (AM^{3}Net), coupled adversarial learning-based classification (CALC), cross-channel reconstruction module (CCR-Net), coupled CNNs (CNN-DF-M), deep encoder–decoder network (EndNet), global–local transformer network (GLT-Net), interleaving perception CNN (IP-CNN), and spatial–spectral cross-modal enhancement network (S^{2}Enet).

Almost all of them use deep CNN architectures, where CCR-Net, IP-CNN, S^{2}Enet, and EndNet are feature-level

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TABLE XI
QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND KAPPA ON THE HOUSTON DATASET. THE BEST ONE IS SHOWN IN BOLD

No.	Class Name	Performance								
		AM^{3}Net	CALC	CCR-Net	CNN-DF-M	EndNet	GLT-Net	IP-CNN	S^{2}Enet	AMSSE-Net
1	Health grass	80.53	80.34	82.53	**83.10**	82.72	82.91	80.91	83.10	**83.10**
2	Stressed grass	**85.15**	83.65	85.06	**85.15**	79.42	84.68	83.08	85.15	**85.15**
3	Synthetic grass	94.85	89.31	98.81	99.60	**100.00**	99.01	70.50	100.00	83.56
4	Trees	86.55	**95.08**	92.90	93.37	91.86	94.03	89.77	92.33	93.37
5	Soil	**100.00**	99.62	100.00	**100.00**	99.24	98.77	97.25	100.00	**100.00**
6	Water	**100.00**	**100.00**	93.01	**100.00**	95.10	95.80	95.80	100.00	93.01
7	Residential	86.47	86.85	86.38	89.65	84.42	90.30	86.38	87.78	**95.34**
8	Commercial	69.23	85.00	76.73	91.45	76.26	77.87	69.23	79.20	**91.74**
9	Road	87.44	86.78	84.99	83.00	72.14	92.35	80.26	90.27	**94.24**
10	Highway	63.80	68.05	63.32	68.05	54.83	64.09	47.10	53.28	**86.20**
11	Railway	93.93	**99.53**	83.11	95.35	84.54	78.56	70.78	82.92	96.11
12	Parking lot 1	92.70	97.31	88.76	94.24	96.93	98.17	89.24	86.65	**99.62**
13	Parking lot 2	84.21	**92.28**	91.23	89.82	71.93	89.12	82.81	91.93	81.40
14	Tennis court	**100.00**	99.60	100.00	98.79	**100.00**	97.98	81.78	100.00	**100.00**
15	Running track	**100.00**	96.83	98.52	98.94	98.52	**100.00**	97.04	100.00	99.00
	OA	86.12	89.08	86.14	89.62	83.90	87.69	80.09	86.09	**92.26**
	AA	88.32	90.68	88.36	91.37	85.86	89.58	81.46	88.84	**92.10**
	Kappa	0.8493	0.8817	0.8501	0.8875	0.8257	0.8669	0.7852	0.8496	**0.9159**

fusion methods; AM^{3}Net and GLT-Net use decision fusion strategies; and CNN-DF-M and CALC use both feature-level and decision-level fusion methods. For a fair comparison, we use precisely the same training and testing samples, and finally compare the results of independent experiments.

For the Houston dataset, which contains 15 types of ground objects, some areas are obscured by clouds and fog, resulting in high scene complexity. We can draw the following conclusions from Table XI. Firstly, for models with feature-level fusion, feature extraction is particularly important before the fusion of multisource data. IP-CNN utilizes the Gram matrix for multisource complementary information preservation to achieve complementary structure preservation and fusion information integrity of HSI and LiDAR sources, but this also leads to the existence of a large amount of redundancy information, which makes it difficult to extract the key information, and finally achieves a more general classification result. EndNet, which is also a feature-level fusion, fuses multimodal information by forcing the fusion of features to sequentially reconstruct the multimodal inputs with a simple but effective multimodal DL baseline. This reconstruction strategy enables better activation of cross-modal neurons, takes full advantage of the encoder–decoder module, aims to fuse HSI and LiDAR data in a more compact way, and achieves higher accuracy than IP-CNN with a lightweight network model that achieves an improvement of about 4%. Compared to EndNet, CCR also learns fused features from different RS data sources through a cross-modal reconstruction strategy, where these modalities can exchange information with each other in a more efficient manner, ultimately generating features that are sharper on the edges and structure of the object, resulting in a more detailed and finer-grained appearance of the fused features, achieving an improvement of about 2.2%. All these methods focus more on the feature fusion approach. In contrast, S^{2}Enet focuses more on the information interaction prior to the fusion of the multisource data, enhancing the spatial representation of the hyperspectral data by LiDAR features and the spectral representation of the LiDAR data by hyperspectral features, which ultimately achieves a better classification performance than the above-mentioned methods. And for some specific areas, especially in the cloud-obscured regions, this method achieves the accurate classification of the categories. As observed from the experimental results, it achieves 100% classification accuracy in five categories, which confirms the effectiveness of the method for the fusion of the HSI spatial–spectral information with the elevation information of the LiDAR data. Secondly, a suitable decision-level fusion strategy helps to improve further the classification performance, AM^{3}Net takes advantage of the feature that shallow and deep appearance features have a strong complementary relationship in representing global and local information, and fuses the three hierarchical features through different weights, similarly, GLT-Net's multiclass estimation based on local and global features is integrated through a probabilistic decision-making fusion strategy, and both achieve better experimental results, with different degrees of improvement over the feature-level fusion approach. Thirdly, using both feature-level and decision-level fusion approaches, CNN-DF-M and CALC adopt this approach to integrate these heterogeneous features in the fusion stage fully, for example, designing weight sharing can simultaneously extract similar higher-order semantic information, perform feature fusion on them, and finally further improve the classification performance through adaptive probabilistic fusion strategy. The experimental results show that the classification performance of this approach is more obviously improved compared to other methods, which is about 3% better than the best method mentioned above, and the OA both reach more than 89%. Therefore, we have considered the advantages of the above methods. Before feature fusion, we use involution to mine the deep spectral features of HSI and then adopt MHF to capture the spatial information so that the feature maps with smaller receptive fields can better capture the detailed information and the feature maps with larger receptive fields can capture the more macroscopic structural and semantic information, thus

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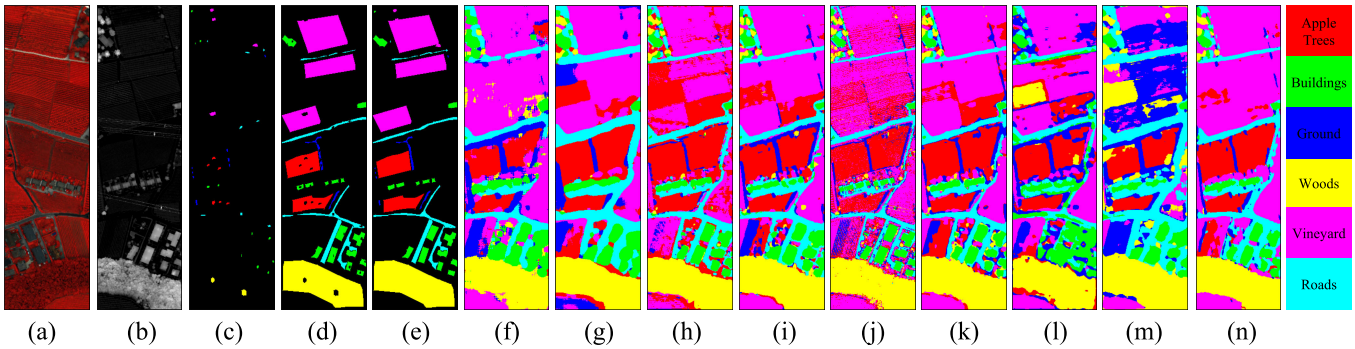


Fig. 9. Visualization and classification maps for the Trento dataset. (a) False-color HSI (using bands 40, 20, and 10, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Training image. (d) Testing image. (e) Ground-truth map. (f) AM^{3}Net. (g) CALC. (h) CCR-Net. (i) CNN-DF-M. (j) EndNet. (k) GLT-Net. (l) IP-CNN. (m) S^{2}Enet. (n) AMSSE-Net.

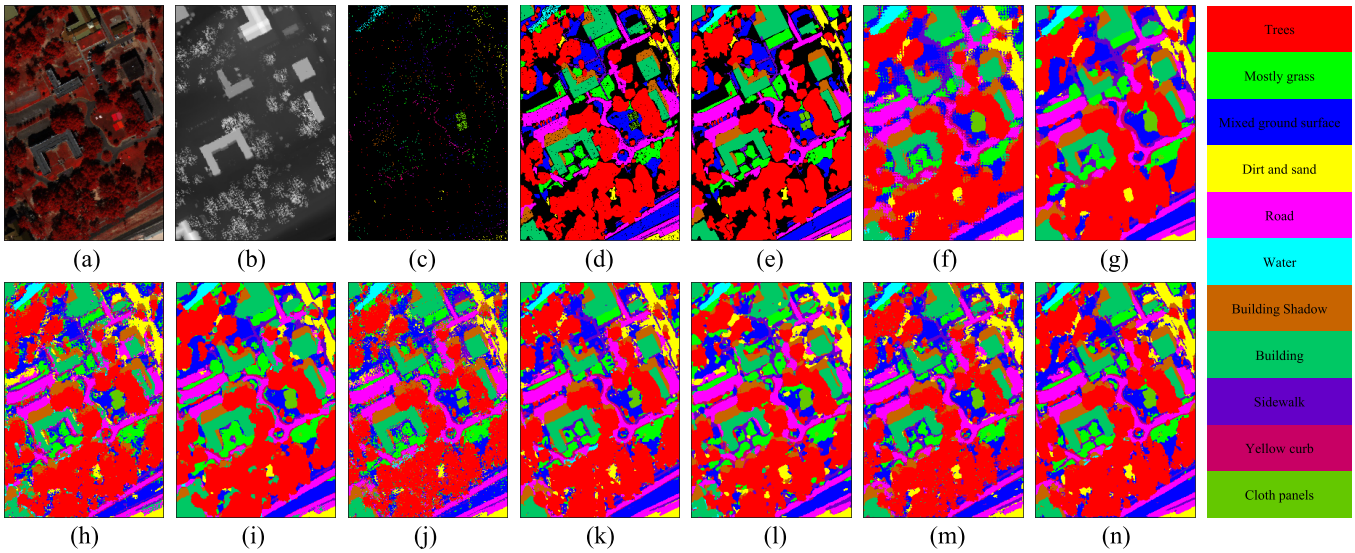


Fig. 10. Visualization and classification maps for the MUUFL dataset. (a) False-color HSI (using bands 40, 20, and 10, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Training image. (d) Testing image. (e) Ground-truth map. (f) AM^{3}Net. (g) CALC. (h) CCR-Net. (i) CNN-DF-M. (j) EndNet. (k) GLT-Net. (l) IP-CNN. (m) S^{2}Enet. (n) AMSSE-Net.

forming a richer and more comprehensive classification performance than other methods. Finally, the data is fused based on the combination strategy of feature level and weighted fusion, which realizes the effective fusion of the spatial–spectral information of HSI and the elevation information of LiDAR data and improves the classification effect of the model. Experimental results show that our AMSSE-Net method achieves the best classification performance on the Houston dataset for OA, AA, and Kappa and the highest classification accuracy in 9 out of 15 categories. For Trento and MUUFL datasets, which have higher spatial resolution and fewer classification categories, the performance of the different classification methods shows a similar trend, which also illustrates the strong generalization of our proposed model.

Table IX presents the quantitative classification results for the Trento data. AMSSE-Net achieves the best performance on the two target metrics, OA and Kappa, with OA more than 0.3% higher than the suboptimal model, and achieves the highest level in two of the six land object categories. Table X lists the quantitative classification results for the MUUFL data, where our method achieves the best performance on the two target metrics of OA and Kappa on the MUUFL data and achieves the highest level of accuracy on the two ground objects.

2) Visual Comparison: Figs. 9–11 sequentially show the hyperspectral false-color images, visualized LiDAR data, training set images, test set images, ground-truth maps, and plots of the visual classification results of the different methods on the Trento, MUUFL, and Houston datasets, with the colors for each category also illustrated. The visualization of the classification results of our proposed method, AMSSE-Net, is closer to the ground-truth map.

For the Trento data, the classification results obtained are smoother and have fewer isolated points compared to the classification maps obtained by other methods, which are basically consistent with the true value maps due to the better information fusion ability. On the MUUFL dataset, our approach efficiently extracts deeper spectral features from hyperspectral images, thereby enhancing its discriminative capability for mixed landmasses. This leads to a clearer and more accurate delineation of individual classes across the entire area, resulting in significantly reduced noise in the generated results. On the Houston dataset, some areas are obviously obscured by clouds and fog, so this part of the area is more testing for the model to utilize the LiDAR data to complement the elevation information of the HSI data. It is clear that the visual classification maps, the resultant maps of each model, have different classifications of fuzzy areas in the

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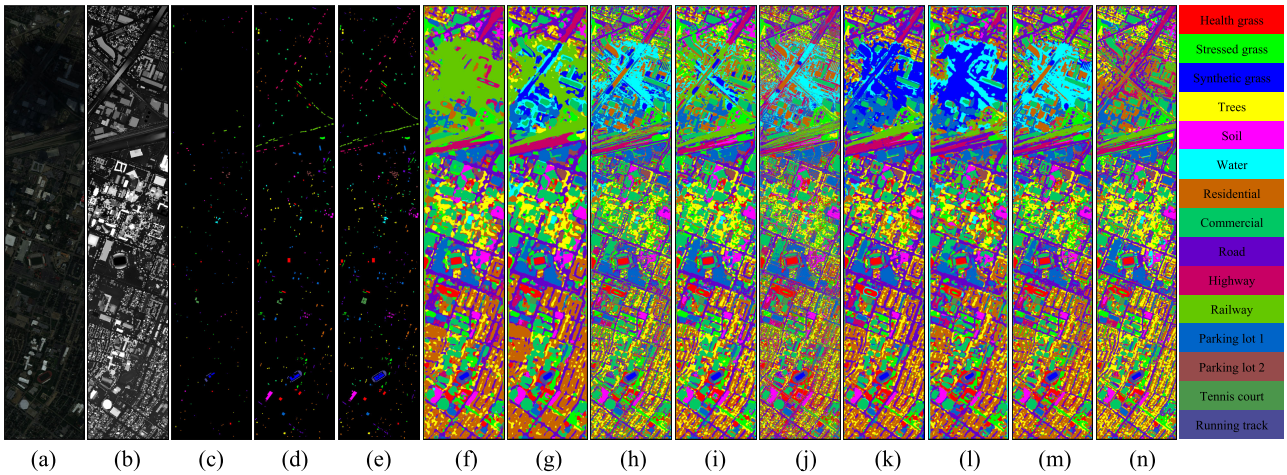


Fig. 11. Visualization and classification maps for the Houston dataset. (a) False-color HSI (using bands 64, 43, and 22, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Training image. (d) Testing image. (e) Ground-truth map. (f) AM^{3}Net. (g) CALC. (h) CCR-Net. (i) CNN-DF-M. (j) EndNet. (k) GLT-Net. (l) IP-CNN. (m) S^{2}Enet. (n) AMSSE-Net.

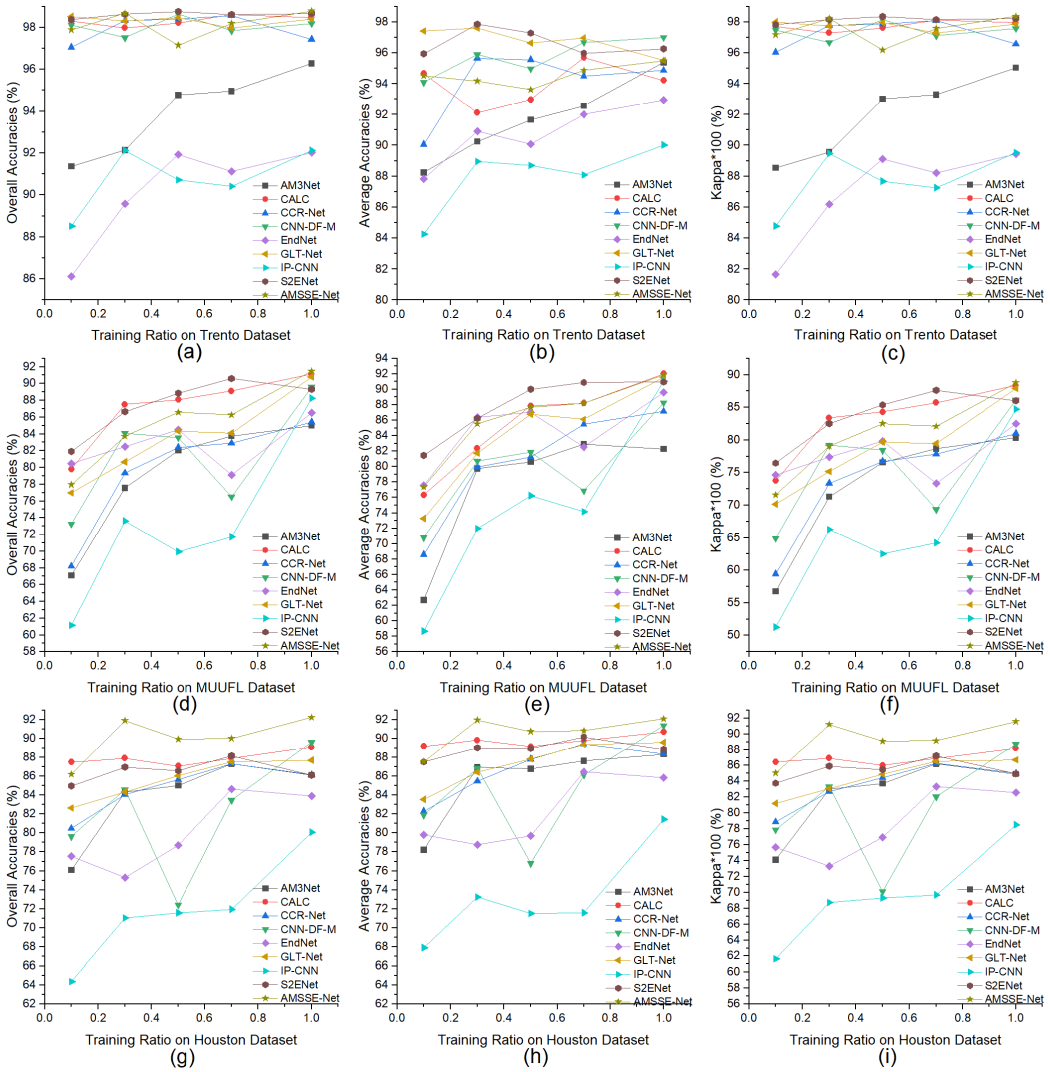


Fig. 12. Impact of training ratio based on the original training samples selection in terms of OA (%), AA (%), and Kappa×100(%) for different methods for three datasets. (a)–(c) OA, AA, and Kappa on the Trento dataset, respectively. (d)–(f) OA, AA, and Kappa on the MUUFL dataset, respectively. (g)–(i) OA, AA, and Kappa on the Houston dataset, respectively.

railway and commercial categories, which are very different, while the classification results of our proposed AMSSE-Net approach not only accurately show the texture in this part, but also accurately distinguish the categories.

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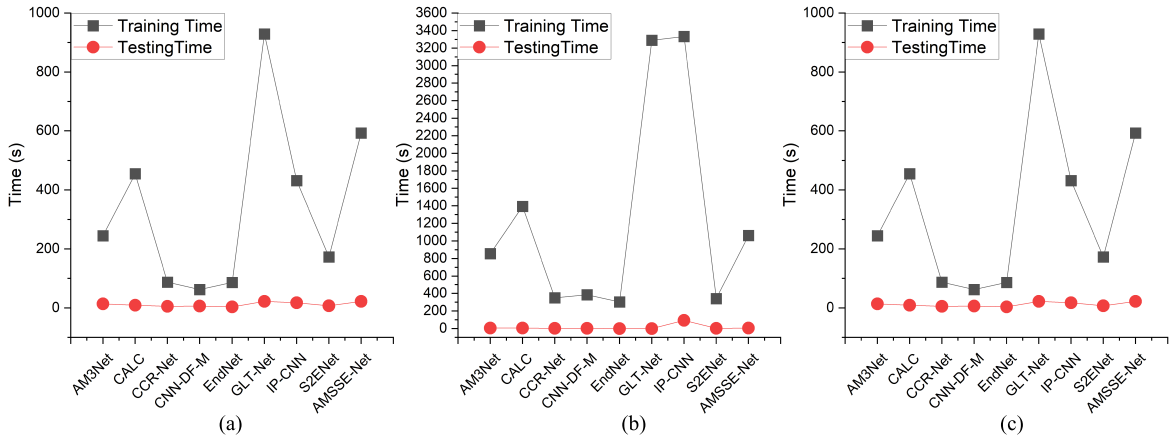


Fig. 13. Comparison of training and testing time(s) of different models for three datasets. (a) Training and testing time on the Trento dataset. (b) Training and testing time on MUUFL dataset. (c) Training and testing time on the Houston dataset.

In conclusion, our AMSSE-Net model and some of the currently proposed models show large advantages in objective and visual comparisons.

3) Performance Comparison for Different Training Ratio: Fig. 12 illustrates the effect of different training samples on the performance of each model, depicted in terms of OA, AA, and Kappa. We gradually set the proportion of training samples (10%, 30%, 50%, 70%, 100%) and investigate the trend of the overall classification accuracy.

As can be seen from the figure, on both the Trento and MUUFL datasets, most of the models are able to achieve high levels of performance with different training samples, but our proposed model, although being in the middle of the range in the case of small samples, has been showing an increasing trend of OA, AA, and Kappa as a whole with the increase of sample sizes, especially on the MUUFL dataset, which eventually reaches the best classification performance among all models at 100% training samples, achieving the best classification performance among all models, which reflects the positive impact of larger number of training samples on our proposed classification method. On the Houston dataset, it is clear that except for the case of 10% samples, where the classification performance of our proposed method is slightly lower than that of CALC, our AMSSE-Net always achieves the best performance when the number of training samples is the same in all other cases, which fully demonstrates the advantage of AMSSE-Net on datasets with a large number of spectral bands such as Houston, which is able to mine the deep spectral information, and also reflects the full interaction of adaptive feature fusion for multimodal information.

E. Computational Complexity Analysis

In order to quantitatively analyze the computational cost of the different models, Fig. 13 shows the training time and testing time of the different models on Trento, MUUFL, and Houston data, respectively. From the figure, it can be found that the training time of our proposed model is at the medium level. From the figure, we can find that the training time and testing time of our proposed model on each dataset are ranked in the middle of the range, but there is not much difference between the time we spend and the time spent by the previous models. It is clear that EndNet, a lightweight network, has a very short training and testing time. In contrast, IP-CNN generates a large number of parameters because of the two stages of training, resulting in a long time for both training and testing. Both our proposed AMSSE-Net and AM^{3}Net use the involution operator to deep mine the spectral features of HSI, and the computation time of the network mainly lies in this part as well. However, the training time and testing time of our network and AM^{3}Net are similar due to the existence of three fusion methods in our proposed AFFM. The AMSSE-Net's training time and testing time are in the acceptable range, yet the best classification results can be achieved, which explains the relatively high training and testing efficiency of our proposed network.

IV. Conclusion

This article proposes an adaptive spatial–spectral enhancement network for classification of hyperspectral and LiDAR data, called AMSSE-Net. Specifically, using the involution module and MHFM to mine deep spectral–spatial features of HSI, the involution operator deeply mines the spectral features of HSI and improves the spectral dimensionality correlation, and at the same time, enables the neurons to be more flexible in capturing the texture, spatial structure, and multiscale features in the input data, spatial structure, and multiscale features; then extract the multiscale information, capture the detail information through the feature maps with smaller receptive fields, and capture the macroscopic structural and semantic information through the feature maps with larger receptive fields, mine the comprehensive spatial information of HSI, and then finally form the spectral–spatial features of HSI with the strategy of hierarchical fusion; finally, fusion classification is performed on the data based on the combination of the feature level and weighted fusion strategy with the use of the self-adaptive weighting, which dynamically assigns weights according to the different contribution degrees of the three outputs to find the optimal model with the joint loss as a constraint. To demonstrate the effectiveness of the proposed AMSSE-Net, we have conducted rich experiments on the HSI and LiDAR datasets of Trento, MUUFL, and Houston, which proved that our proposed method outperforms other state-of-the-art methods. In summation, our research provides a foundational framework for advancing the field of multimodal remote sensing data analysis and classification.

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We can further explore the generalization of the AMSSE-Net architecture to handle different types of remote sensing data. Additionally, the development of more efficient and adaptive fusion mechanisms for multimodal data remains a key area of interest. The further exploration of adaptive fusion mechanisms can enhance the model's applicability in diverse remote sensing applications. Overall, our work lays the foundation for continued advancements in the field of multimodal remote sensing data analysis and classification. However, there are still some limitations, and the mining of Lidar data is insufficient. In future research, more attention can be paid to the information interaction before multimodal feature fusion, which is the focus of our future work.

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